

SHADE MATSUMOTO

Honolulu, HI | Willing to Relocate

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Education

University of Hawaii at Manoa

Aug 2023 – Dec 2027

BS Computer Science | 3.94 GPA

Honolulu, HI

Experience

Data Science Intern

Aug 2026 – Present

US Indo-Pacific Command

Honolulu, HI

- Developed AI/ML-enabled analytics and decision-support solutions using large-scale enterprise data.
- Designed enterprise data integration pipelines and modernization initiatives to improve operational efficiency.

Undergraduate Research Assistant

Jul 2026 – Present

Lab for Advanced Visualization & Applications

Honolulu, HI

- Designed C++ software for interactive scientific visualization, transforming complex multidimensional datasets into responsive 2D/3D representations for large-scale and immersive display systems.
- Implemented real-time rendering and data-processing pipelines, optimizing memory usage, frame performance, and user interaction for efficient exploration of large research datasets.

Software Engineering Intern

Jun 2025 – Aug 2025

NOIRLab Gemini Observatory

Hilo, HI

- Engineered physics simulation of international telescope to model mass-balancing structure in Unreal Engine, enabling virtual testing vs manual calculation, reducing part-replacement-operation hours by estimated **50%+**.
- Built C++ and Python backend pipelines with Unreal Engine APIs to support object data access and asset imports.
- Optimized simulation efficiency by tuning Nanite parameters to manage high-fidelity 3D assemblies, cutting frame latency by **15%** and VRAM footprint by **30%** compared to UE5 base parameters.
- Developed real-time interface for simulation, collaborating with engineers to rapidly prototype features.

Operations Center Student Assistant

Oct 2024 – Aug 2026

University of Hawaii IT Services

Honolulu, HI

- Resolved **800+** software, hardware, and network issues for university members through desk-side support and tickets.
- Maintained operational health of infrastructure and data center equipment through routine monitoring & diagnostics.

Projects

AI Image Recognition | PyTorch, Python

Apr 2025

- Implemented backend services to ingest image uploads alongside parsed EXIF metadata and temperature logs.
- Trained a PyTorch convolutional neural network for cloud type classification, achieving **92%** accuracy.
- Deployed the model via Gemini Enterprise Agent Platform, building a real-time inference pipeline.

Campus Web App | Next.js, PostgreSQL

May 2025

- Developed web application using React Bootstrap and Next.js to centralize campus food vendor data and schedules.
- Integrated REST API routes with Node.js and TypeScript to query a PostgreSQL database.
- Normalized public vendor data into unified JSON schema, automating migration of legacy data into relational backend.
- Deployed the full-stack app to Vercel using GitHub Actions for CI/CD.

Skills

Languages: C++, Python, Java, C#, TypeScript, JavaScript, C, SQL, Bash

Frameworks: PyTorch, React, Next.js, Node.js, NumPy, JUnit

Databases: PostgreSQL, MySQL, Redis

DevOps: Docker, Kubernetes, AWS, Google Cloud, GitHub Actions, CI/CD, Cloud Computing

Platforms: Linux, Git, Unreal Engine, Gemini Enterprise Agent Platform

Awards

Booz Allen Hamilton Scholarship (2026)

Manoa Academic Merit Scholarship (2023 – Present)